

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Dept. of E&ECE

2. Subject Name: DSP LABORATORY L-T-P: 0-0-3 Credits: 2

3. Pre-requisites: -

4. Syllabus and reference books:

Sampling theorem: Sampling above and below Nyquist rate. Effect of aliasing. Spectrum of a square wave. Interpolation and upsampling. Designing low pass filters by windowing method: Implementation and comparison of rectangular window, triangular window, Hanning window, Hamming window, Blackman window. Dual tone multi-frequency (DTMF) encoder/decoder. Power spectrum estimation: The Welch nonparametric method. Parametric methods, The Yule-walker AR model. Adaptive line enhancer. Filter design using *simulink* toolboxes.

Reference Books:

1. *Digital Signal Processing*, by Oppenheim and Schaffer.
2. *Digital Signal Processing*, by J. G. Proakis and D. G. Manolakis

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Sampling theorem.	2 (6 hours)
2.	Designing low pass filters by windowing method	2 (6 hours)
3.	Dual tone multi-frequency (DTMF) encoder/decoder	2 (6 hours)
4.	Power spectrum estimation	2 (6 hours)
5.	Adaptive Line enhancer	2 (6 hours)
6.	Filter design using <i>simulink</i> toolboxes.	1 (3 hours)
Total number of hours		11 sessions.